

# Phaser

A stereo phase shifter with three voicings: the orange stompbox, the slow opto and real tape phasing.



## Overview

Phaser runs your signal through a chain of all-pass stages and mixes it back with the dry sound. Where the two cancel you get notches, and as the LFO moves the stages those notches sweep up and down the spectrum - the classic swirling, liquid phase sound.

Three voicings are built in. 90 is the little orange stompbox: a focused, vocal sweep with its peaks gently rounded off. OPTO follows the old lamp-and-photocell units: the sweep rises quickly and falls back slowly, so it sounds rounder and deeper over a wider range. TAPE is a different machine altogether - phasing done with a short delay, as with two tape machines, including a hand-made motion that wanders like a thumb on the reel.

The rate can run free, lock to the project tempo, or follow the tempo while flowing through loops. A DRIFT dial adds slow, random wander to the rate, depth and centre so the effect never sounds mechanical.

## Reach for it for

- Swirling rhythm guitar and electric piano
- Slow, deep sweeps on pads and synths
- Tape-style phasing on drums, a bus or a whole mix
- Tempo-locked throb with the square LFO

- Wide stereo movement with SPREAD

## Controls

### VOICING (header)

|             |                                                                                                                                                                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>90</b>   | The orange stompbox. The sweep covers about 120 Hz to 2.4 kHz, and peaks above about -3 dBFS are gently rounded off on the way out. With this voicing a SCRIPT / BLOCK key appears beside the rate keys (see below).                                                                   |
| <b>OPTO</b> | The lamp-and-photocell phaser. The sweep rises fast and falls back slowly, covers a wider range (about 80 Hz to 2 kHz) and sounds smoother and deeper.                                                                                                                                 |
| <b>TAPE</b> | Phasing with a swept short delay (about 0.6 to 8 ms at its centre) against a slightly delayed dry signal, as with two tape machines. The swept copy can pass right through the dry one for the deep cancellation of real tape phasing. In this voicing the STAGES knob becomes MOTION. |

### Display

The window on the left shows the live frequency response - the notches moving as the effect sweeps. The top line shows the number of stages and the current sweep frequency, or in TAPE the current delay time in ms.

### LFO SHAPE and RATE MODE

|                                 |                                                                                                                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SINE / TRI / SQR / RND</b>   | The LFO shape. Triangle is the even, classic sweep; sine lingers at the ends; square jumps between two settings for a throbbing effect; random wanders from point to point. |
| <b>FREE / SYNC / FLOW</b>       | How the rate is set: FREE in Hz, SYNC locked to the project tempo and timeline, FLOW tempo-synced but running continuously through loops.                                   |
| <b>SCRIPT / BLOCK (90 only)</b> | A quick feedback switch for the 90 voicing. SCRIPT is the original with no feedback; BLOCK adds 40% feedback for a stronger, more pronounced sweep. If FEEDBACK has been    |

|                           |                                                                                                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | set to any other value the key reads CUSTOM; clicking it then returns to SCRIPT.                                                                                                        |
| <b>STAGES</b>             | The number of all-pass stages: 2, 4, 6, 8 or 12. More stages give more notches and a thicker, more complex sweep; 4 is the classic pedal. Click a number to pick it. Not shown in TAPE. |
| <b>MOTION (TAPE only)</b> | What moves the tape: LFO (the rate and shape above), HAND (a slow, random wander like a hand on the reel) or BOTH (the two blended).                                                    |

## Phaser

|                 |                                                                                                                                                                                                                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RATE</b>     | The LFO speed, 0.02 to 10 Hz, or a note value from 4 bars to 1/32 (with dotted and triplet values) when SYNC or FLOW is on. The light beside it blinks with the LFO. In TAPE with MOTION set to HAND the knob becomes HAND SPEED: how fast the hand-made wander moves (0.02 to 10 Hz). |
| <b>CENTRE</b>   | Where the sweep sits, 0 to 100%. Low settings put the notches low in the spectrum; high settings move them up. In TAPE it sets the centre delay time.                                                                                                                                  |
| <b>DEPTH</b>    | How far the sweep travels either side of CENTRE, 0 to 100%. Zero gives a static notch filter.                                                                                                                                                                                          |
| <b>FEEDBACK</b> | Feeds the output back into the stages, from -95% to +95%. Positive sharpens the notches into a ringing, vocal tone; negative gives a hollower colour. The level is compensated so strong settings don't jump out.                                                                      |
| <b>SPREAD</b>   | Offsets the right side's sweep from the left, from 0 to 180 degrees. 0 is a classic mono phaser; 90 - 180 sweeps the two sides against each other for wide stereo motion. In TAPE with HAND motion it sets how independently the two sides wander.                                     |

## Drift

A slow random wander that keeps the sweep from sounding mechanical.

**DRIFT**

0 to 100%. 50% is the standard amount; 100% is twice that, 0 turns it off. It moves the LFO rate a little, and the depth and centre of the sweep a little more. Higher settings also make the wander move faster.

## IN / OUT

**TRIM**

Input level before anything else, from -24 to +12 dB.

**GUARD**

Click the TRIM knob (without turning it) to switch GUARD on or off; the small light at the knob's upper right shows it is on and glows brighter while it works. On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources.

**SAT**

Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.

## Master

**MIX**

Dry / effect balance, 0 to 100%. 50% gives the deepest notches; lower is subtler; at 100% you hear only the shifted signal, which gives a gentle pitch wobble (vibrato) rather than phasing.

**LOW CUT**

A high-pass filter on the effected signal only, from off (fully left) up to 500 Hz. The bass stays dry, clean and centred while the mids and highs swirl.

**OUTPUT**

Output level, -24 to +12 dB.

**OUT meters**

Output level, left and right.

## The header

|                                     |                                                                                                                                                                                                                                                                           |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PRESETS, ◀ ▶ and the display</b> | PRESETS (or a click on the display) opens the list of factory presets; ◀ and ▶ step through them, and the display shows the number and name of the preset in use. Choosing one sets the plugin's controls for that job; the input and output stages are left as they are. |
| <b>S / M / L</b>                    | Panel size: small, medium or large. The panel also scales to fit the plugin window.                                                                                                                                                                                       |
| <b>?</b>                            | Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.                                                                                                                                                |

## Presets

|                          |                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------|
| <b>90 Script</b>         | The classic orange pedal: 4 stages, triangle sweep at 0.6 Hz, no feedback.              |
| <b>90 Block</b>          | The same pedal with 40% feedback for a stronger, more vocal sweep.                      |
| <b>90 Slow Swell</b>     | The 90 voicing with a slow (0.15 Hz), deep sweep.                                       |
| <b>Opto Lush</b>         | OPTO, 8 stages, slow sine sweep with some feedback and the right side 90 degrees apart. |
| <b>Opto Square Throb</b> | OPTO, 6 stages, square LFO locked to 1/8 notes for a rhythmic throb.                    |
| <b>Opto Stereo Swirl</b> | OPTO, 12 stages, strong feedback and the two sides 180 degrees apart.                   |
| <b>Opto Vibrato</b>      | OPTO at 4.5 Hz with MIX at 100%: a pitch vibrato instead of phasing.                    |
| <b>Hand Tape Phasing</b> | TAPE moved by hand: a slow, uneven, wandering sweep.                                    |
| <b>Tape Jet</b>          | TAPE with 60% feedback and both LFO and hand motion, for the jet-plane sound.           |

|                              |                                                                                              |
|------------------------------|----------------------------------------------------------------------------------------------|
| <b>Through-Zero Sweep</b>    | TAPE with a very slow, full-depth sweep that passes right through the dry signal; drift off. |
| <b>Reset all to defaults</b> | Every control back to its starting value, including the input and output stages.             |

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## Tips

- For the classic tape-phasing effect on a mix bus, use TAPE with FEEDBACK at 0, a slow sine sweep and MIX at 50%.
- On bass or a full mix, raise LOW CUT to 100 - 200 Hz so the low end stays solid while the top swirls.
- Try the square LFO with SYNC on for a rhythmic, tremolo-like throb that sits in time with the track.
- If the effect sounds too thin or nasal, lower FEEDBACK or move CENTRE down; for more drama, add stages.

File: gfx-phaser.jsfx